

Title: DPT

Date:

Model

$$W[A_{ij}|0 \rightarrow 1] = \frac{\alpha}{N} + \beta(A^2)_{ij} \quad W[A_{ij}|1 \rightarrow 0] = r$$

$$\text{平均度} \quad \frac{dz}{dt} = \alpha - rz + \beta(z^2 - \Delta)$$

$$\frac{dz}{dt} = \alpha(1 - \frac{z}{N}) - rz + \frac{1}{N} \langle \sum_{ijk} A_{ik} A_{kj} (1 - A_{ij}) \rangle$$

$$\beta = \frac{1}{z^2 - \Delta}$$

$$\Delta: \text{三元组} \quad \frac{d\Delta}{dt} = \beta(z^2 - \Delta) - 3r\Delta + O(\beta^3) \quad (\vec{1} - \vec{r})$$

$$z(t) \sim \begin{cases} \text{constant} & (\beta \leq \frac{r^2}{4\alpha}, t \rightarrow \infty) \\ \frac{1}{t - t_c} & (\beta > \frac{r^2}{4\alpha}, t \rightarrow t_c) \end{cases}$$

After DPT

p : iso node, $(1-p(t))N$: complete graph

$$q(t) = (1-p(t))^2 \quad dp/dt = \alpha t, \quad p(t) = p_0 e^{-\alpha(t-t_c)}$$

adiabatic approx:

$$\frac{d\Delta}{dt} = 0, \quad \Delta = \frac{\beta z^2}{\beta + 3r}, \quad \beta' = \beta(1 - \frac{\beta}{\beta + 3r})$$

after the DPT:

$$\text{core } (1-p(t))N, \quad q(t) = \frac{z}{N} = (1-p(t))^2$$

$$dp/dt = \alpha p, \quad p = e^{-\alpha(t-t_c)} p_0$$

Title: DPT

Date:

相变时间

$$t_c = w^{-1}[\text{acot}(2w) + \pi/2] \quad w = \sqrt{\alpha\beta - r^2/4}$$

$$\frac{dz}{dt} = \alpha - rz + \beta z^2 \quad \beta(z - \frac{r}{2\beta})^2 + \alpha - \frac{r^2}{4\beta}$$

$$\frac{1}{pu^2 + q} du = dt \quad u = z - \frac{r}{2\beta}, \quad p = \beta, \quad q = \alpha - \frac{r^2}{4\beta}$$

$$\frac{1}{\sqrt{pq}} \arctan \sqrt{\frac{p}{q}} u = t + C$$

$$t=0, z=0, u = -\frac{r}{2\beta} \quad C = \frac{1}{\sqrt{pq}} \arctan -\frac{r}{2\beta} \sqrt{\frac{p}{q}} \quad pq = \alpha\beta - \frac{r^2}{4} = w^2$$

$$\sqrt{pq}(t_c + C) = \frac{\pi}{2}$$

$$t_c = \frac{1}{\sqrt{pq}} \left(\frac{\pi}{2} - \arctan -\frac{r}{2\beta} \sqrt{\frac{p}{q}} \right)$$

$$p/q = 4\beta^2 / (4\alpha\beta - r^2)$$

$$w^{-1}(\text{acot}(2w) + \pi/2)$$

$$t_c = w^{-1} \left(\frac{\pi}{2} + \text{acot} \frac{\sqrt{4\alpha\beta - r^2}}{2w/r} \right)$$

临界行为

$$\frac{\partial P(k,t)}{\partial t} = (\alpha + \beta(k-1)z(t))P(k-1,t-1) + r(k+1)P(k+1,t-1)$$

$$-(\alpha + \beta k z(t) + rk)P(k,t-1)$$

$$\frac{\partial G}{\partial t} = (\alpha X - \alpha)G + (\beta Z X^2 + r - \beta Z X - rX) \frac{\partial G}{\partial X}$$

$$= (X-1) \left[\alpha G + (\beta Z X - r) \frac{\partial G}{\partial X} \right]$$